

Upgrading to Gen7/Gen7-DL Image Compute Node (ICN) on a Discovery MR750, MR750w, or Signa Architect System

Gen7/Gen7-DL Upgrade Manual



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Language Policy

Direction 2128126 - Language Policy For Service Documentation

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警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obsluženého personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAAR-SCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlke-teenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. - Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. - Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ÆDVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploataavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploataavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCIÓN (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ЗАСТЕРЕЖЕННЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. - Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. - Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. - Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.

Revision history

Record of reason for changes to this document.

Rev	Date	Description
5	January 2022	Routed in MyWorkshop with DOC2270583, revision 6. This revision adds support for the Dolphin 2 interface connection. The following updates were made: - Chapter 5, Section 5.2 Installing the ICN: - Added new step 1 - Added note to step 2 about changing between ICN styles - In Figure 5-2ICN cable connections (Gen 7 and Gen 7-DL, Dell R640), updated description in callout 1 - Added Figure 5-3 PCIe cable connection - Dolphin 2 card - Chapter 7, Section 7.1 Determining software installation: - Updated Table 7-2 Software upgrade steps to include an asterisk in the two SW 29 column headers for the Dolphin 2 footnote reference - Added footnote content for Dolphin 2 card and supported software
4	December 2020	- Updated title to include DV products. This document now applies to DV products as well as Architect. - Chapter 1 1.2.3 Updated harvest test information 1.2.4 Updated harvest list and part numbers 1.4 Updated product locator information 1.6 Added system and component environmental conditions - Chapter 5 5.2 Changed ISC to PGR - Chapter 7 7.1 Updated software upgrade steps table to include Rev 5 and earlier ICNs Revision 4 is referenced as DOC2270583, rev 5, in MyWorkshop.

Rev	Date	Description
3	June 2020	- Chapter 1 1.2.4 Added harvest ICN options and part numbers 1.3 Removed information not relevant to this manual 1.7 Removed non-magnetic tools - Chapter 5 - Added Replacing the Image Compute Node (ICN) network cables topic 5.2 Clarified installation steps and added Gen 7-DL relevant cabling information. - Chapter 7 - Changed chapter 7 to Determining software installation - Chapter 8 - Added Gen 7-DL and MR29 SW relevant information - Chapter 9 - Removed Logging in to the Image Compute Node (ICN) chapter Revision 3 is referenced as DOC2270583, rev 4, in MyWorkshop.
2	Nov 2019	Correct erroneous manual number on front page and footer throughout entire manual. Revision 2 is referenced as DOC2270583, rev 3, in MyWorkshop.
1	July 2019	Initial Release of part number 5829487. Revision 1 is referenced as DOC2270583, rev 2, in MyWorkshop.

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Contents

Chapter 1 Getting started	13
1.1 General safety precautions	13
1.2 Material policy overview	13
1.2.1 Returned material.....	13
1.2.2 Disposal of material	13
1.2.3 Harvest test procedure.....	13
1.2.4 Harvest items	14
1.3 Product delivery instructions.....	15
1.4 Product locator.....	15
1.5 Damage in transportation.....	16
1.6 MR system shipping and storage requirements.....	17
1.7 Tools and materials	17
Chapter 2 Shutting down the host computer	19
2.1 Shutting down the host computer	19
Chapter 3 Applying LOTO.....	21
3.1 Applying LOTO - PGR PDU/Gradient Subsystem	21
Chapter 4 Removing the Image Compute Node (ICN)	25
4.1 Removing the ICN	25
Chapter 5 Installing the Image Compute Node (ICN)	27
5.1 Replacing the Image Compute Node (ICN) network cables	27
5.2 Installing the ICN	27
Chapter 6 Removing LOTO.....	29
6.1 Removing LOTO - PGR PDU/Gradient Subsystem.....	29
Chapter 7 Determining software installation	31
7.1 Determining software installation	31
Chapter 8 Configuring the Image Compute Node (ICN) - VRE	33
8.1 Configuring the ICN - VRE.....	33
Chapter 9 Doing a check scan.....	39
9.1 Doing a check scan.....	39

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Chapter 1 Getting started

1.1 General safety precautions

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare <i>MR Service Safety Manual</i> (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

1.2 Material policy overview

1.2.1 Returned material

Based on GE Healthcare Policy 3.9, all requested material is to be promptly returned as directed, intact to either GoldSeal or Approved GE Recycling Center. Do not remove or swap any parts from the material to be returned.

1.2.2 Disposal of material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. To make sure you are compliant with all local, state, or national environment, health, and safety regulations, dispose of all items based on the local GEMS Policy and Procedures, and other applicable published guidance. Contact the GE Recycling Center for assistance.

1.2.3 Harvest test procedure



NOTICE

RISK OF DAMAGE

Equipment may be damaged prior to removal.

The equipment must be tested prior to the system shutdown/upgrade to allow appropriate harvest capabilities.

Before going on-site:

- Print the proper regional address labeling from *Harvest Return Addresses* (DOC2100599).
- Make sure you can download the appropriate *System Harvest Test Procedure* from the online documentation library using:
 - DOC1899393 for MR450, MR450W, MR750, MR750W, Artist, and Architect

- If you have not been contacted to test the system or cannot download the document, contact the local harvest team. For all other requests, contact the @HEALTH Global Harvest Team via email: GlobalHarvestTeam@ge.com.

After testing is complete, submit the test record for review. After these test records are reviewed, they will be processed for electronic signatures.

Package the equipment and follow your region's method of returning harvest equipment.

NOTE

Where possible, use packaging from the new upgraded component for the equivalent harvested parts.

1.2.4 Harvest items

Table 1-1 Return the following items for harvest after deinstallation:

Item	Part number
SUN 4100 (Gen2) ICN with VRF	5127452-2
or	or
SUN 4100 (Gen2) ICN with VRF and without Infiniband	5127452-6
or	or
SUN 4170 (Gen3) DV 16-Channel non-GEM	5337894
or	or
SUN 4170 (Gen3) DV 32-Channel non-GEM	5337894-2
or	or
SUN 4170M2 (Gen4) DV 16-Channel non-GEM	5911000
or	or
SUN 4170M2 (Gen4) DV 32-Channel 32 GB non-GEM	5911000-2
or	or
SUN 4170M2 (Gen4) DV 32-Channel 72 GB GEM	5911000-3
or	or
Dell R620 (Gen5) DV 16-Channel ICN with VRF	5921000
or	or
Dell R620 (Gen5) DV 32-Channel ICN with VRF	5921000-2
or	or
Dell R620 (Gen5) DV 32-Channel+ ICN with VRF	5921000-3
or	or
Dell R630 (Gen6) DV Value with Dolphin Card	5931000
or	or
Dell R630 (Gen6) Performance ICN with Dolphin 1 Interface Card	5931000-3
or	or
Dell R630 (Gen6) Advanced ICN with Dolphin 1 Interface Card	5931000-5
or	or

Table 1-1 Return the following items for harvest after deinstallation: (Table continued)

Item	Part number
Dell R640 (Gen7) Performance with Dolphin 1 Card or	5941000-3 or
Dell R640 (Gen7) Advanced ICN with Dolphin 1 Interface Card	5941000-5

1.3 Product delivery instructions

- The *Shipping Document* lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if catalogs and/or packing boxes are missing and/or noted as shipped short.
- **NOTE**
For full systems and large components, two or more people are required to accept delivery of equipment due to weight and size.
- Labels that are attached to the outside of packing boxes summarize box contents.

Product Delivery Instructions (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

1.4 Product locator

During installation, the Install FE shall confirm that the model and serial numbers are part of the asset. Use either the gib web interface (http://gib.gehealthcare.com/gib/gib_entry.jsp) or the MyPLCReader app. This information will then be loaded directly to Siebel by the IB verification team. If components are missing, submit a Global Installation Quality (GIQ) request.

To install the MyPLCReader app, use the following procedure.

1. From the GE app icon, search for and Install the MyPLCReader app on your iPhone or iPad.

Figure 1-1 MyPLCReader app

2. Update the Configuration page.
 - a. The first time the app opens, the Configuration page will open.
 - b. Enter your SSO, Country and **if in US select the edit button to change the default email address** (by default, this app will send the email to the Europe IB Team Resource.EuropeanCentralAdminGIB@ge.com). The email to be used in US is usa.ib.verification@ge.com.
3. Start your first scan.
 - a. Go to the Menu, make the appropriate selection for the job you are performing: **New Install**, **Upgrade System**, or **Update System**.
 - b. Enter the applicable data, install end date, system ID, and RFS number.
 - c. Click **Scan Bar Codes** and start scanning each bar code on the PL Cards.
 - d. Click the **Email** button.
 - e. The email is sent to the default email address, which was set up earlier.

1.5 Damage in transportation

Examine all packages closely at delivery. If damage is apparent, do the following:

1. Make sure the notation *damage in shipment* is written on all copies of the freight or express bill before delivery is accepted or signed for by a General Electric representative or a hospital receiving agent.
2. Whether noted or concealed, you must report damage to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and hold the contents and containers for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

1.6 MR system shipping and storage requirements

NOTE

Some content may not apply to your components.

1. The customer must provide a shipping dock or other staging area for unloading system components for full systems and large components (magnet, cabinets, patient table, and so on).
2. The customer is responsible for unloading and moving all system components from the delivery truck or van to final locations in the MR suite. This includes:
 - Door and other openings are sufficiently wide to allow passage
 - Sufficient space is provided for any required dollies or rigging
3. MR systems and components must be shipped and stored within the environmental requirements as listed in [Table 1-2 Transportation and storage environmental conditions on page 17](#).

Table 1-2 Transportation and storage environmental conditions

System equipment	Temperature range	Temperature change	Relative humidity	Humidity change
Electronic Cabinets and Equipment	-22 to 140°F (-30 to 60°C)	68°F/Hr (20°C/Hr)	10-80% non-condensing	30%/Hr
Surface Coil	-22 to 122°F (-30 to 50°C)	68°F/Hr (20°C/Hr)	10-95% non-condensing	30%/Hr

1.7 Tools and materials

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Brass Master Padlock with Brass Shackle	1	46-194427P320	-
Red LOTO Personal Lock Wrap	2	2393068	-
Red Warning LOTO Tag	1	46-194427P322	-
Digital Voltmeter (DVM)	1	46-194427P284	-

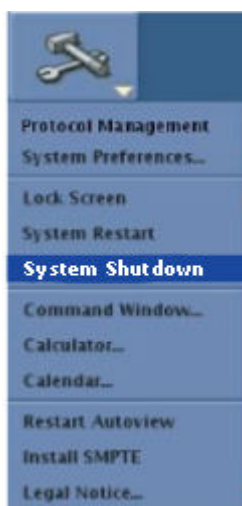
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Chapter 2 Shutting down the host computer

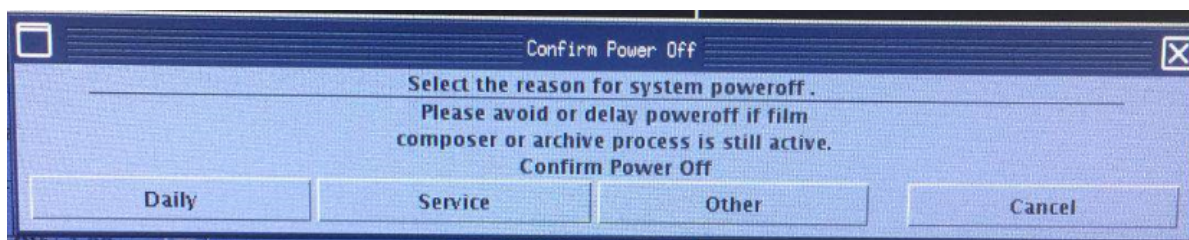
2.1 Shutting down the host computer

1. To shut down the system, click the arrow on the **Tools High Level Access** tab.
2. Click **System Shutdown**.

Figure 2-1 System power off



3. Click **Yes** to confirm the system power off.
4. Select the reason for power off to confirm.



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Chapter 3 Applying LOTO

3.1 Applying LOTO - PGR PDU/Gradient Subsystem



⚠ DANGER

ELECTROCUTION HAZARD!

HIGH VOLTAGE PRESENT.

USE PROPER LOCK OUT / TAG OUT PROCEDURES BEFORE servicing and/or maintaining.

Overview

The PGR cabinet contains the Power Distribution Unit (PDU) and eXtreme Gradient Amplifier (XGA). The XGA is part of the eXtreme Gradient Driver (XGD) subsystem. Prior to servicing and/or maintenance work on the PGR PDU or any element of the XGD, the steps in this Lock Out / Tag Out (LOTO) procedure must be completed by a LOTO-authorized GE Field Engineer (FE). Completing all the steps in this procedure ensures a safe environment and avoids equipment damage when working on these parts.

Table 3-1 Affected by LOTO

Name of Equipment	Number of Locks	Titles of Employees Authorized to perform LOTO	Titles of Affected Employees	How to Notify
- eXtreme Gradient Amplifier (XGA) - eXtreme Power Supply (XPS) - Power, Gradient, RF Cabinet Power Distribution Unit (PGR PDU) - Gradient Filter	1 per GE FE	GE FEs	Hospital personnel	Verbal, posted signs

NOTE

The following table describes the type, location, and magnitude of energy to be LOTO'd.

Energy source	Yes	No	Location of energy isolating means	Magnitude of energy
Electrical	x		PGR PDU circuit breaker panel, Main Disconnect Panel (MDP) circuit breaker panel	208 VAC, 420 VAC, 380-480 VAC
Pneumatic		x		
Hydraulic		x		
Gas/Water/Steam		x		
Chemical		x		
Mechanical Motion		x		
Gravity		x		
Springs		x		

Energy source	Yes	No	Location of energy isolating means	Magnitude of energy
Thermal		x		
Stored Energy	x		Time discharge of internal capacitors	208 VAC, 420 VAC
Air Under Pressure		x		
Oil Under Pressure		x		
Water Under Pressure	x		Heat Exchanger Cabinet (HEC), PGR Pump Circuit Breaker (PPMPCB) and RF amp quick disconnect	5.0 Bar (80 psi)
Gas Under Pressure		x		
Steam		x		
Other		x		

Types of equipment and/or methods selected to dissipate or isolate stored energy

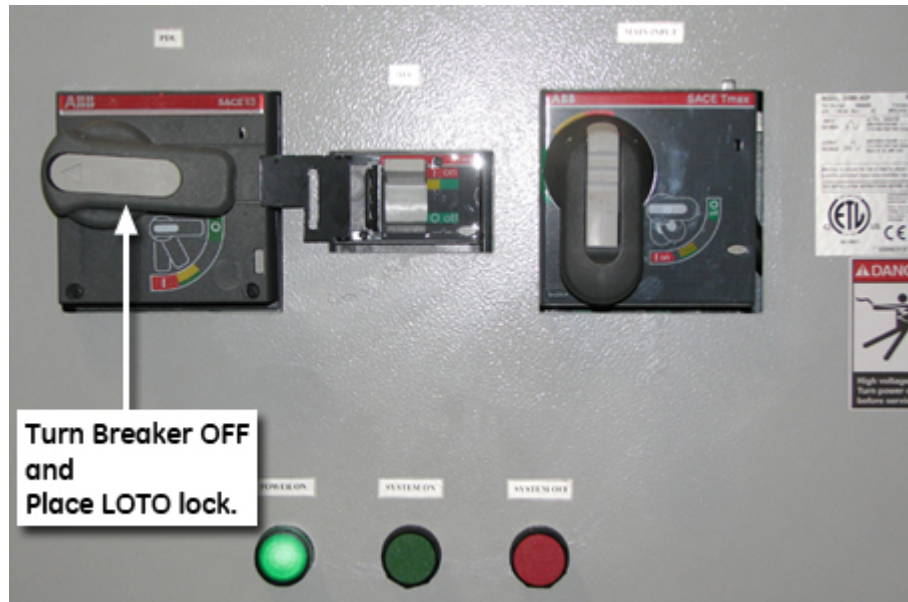
Time method for dissipation of stored energy: 5 minutes.

Type(s) of equipment and/or method(s) used to ensure disconnection(s)

1. One personal lock and tag per FE at the site working on the de-energized equipment at the time of LOTO.
2. Voltmeter to read voltage for verification that energy has dissipated in the PDU.

Applying LOTO

1. To prepare for shutdown of equipment, notify affected personnel working in the area the LOTO is being performed.
2. At the Operator Workspace, shut down the host PC.
3. Locate and turn off the PGR PDU main breaker on the MDP that feeds the PGR cabinet.
Lock Out/Tag Out the PDU breaker on the MDP.

Figure 3-1 PDU breaker on new style MDP

4. Make sure all energy has been dissipated. Use a voltmeter to measure the line-to-line voltage between each INPUT phase voltage test point on the PDU.

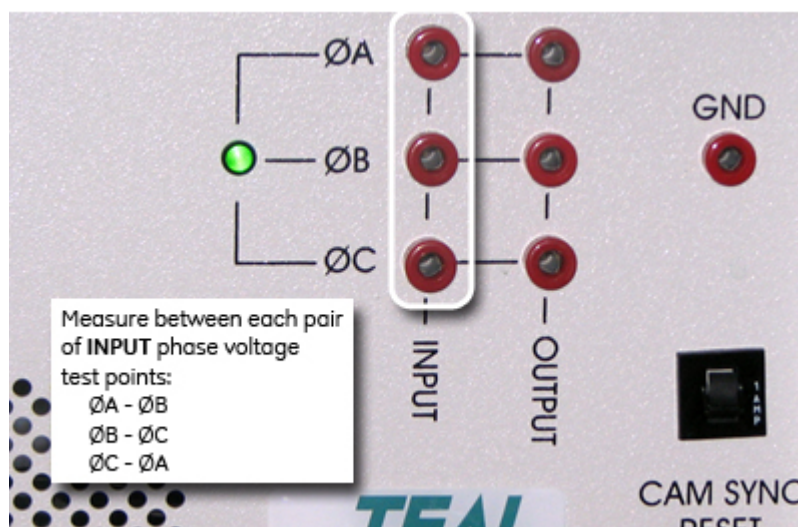
The PDU main breaker must be in the On position to read accurate voltage on the test points.

NOTE

The test points are 100:1 ratio (480 VAC = 4.80 VAC, 208 VAC = 2.08 VAC). Therefore, a low voltage reading still indicates the presence of high voltage. **Your voltage reading for each of the three phases must be 0 VAC before proceeding.**

Input Voltage Range: 380 - 480 VAC = 3.8 - 4.8 VAC

Output Voltage Range: 208 VAC = 2.08 VAC

Figure 3-2 New style PDU test points

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Chapter 4 Removing the Image Compute Node (ICN)

4.1 Removing the ICN

NOTICE

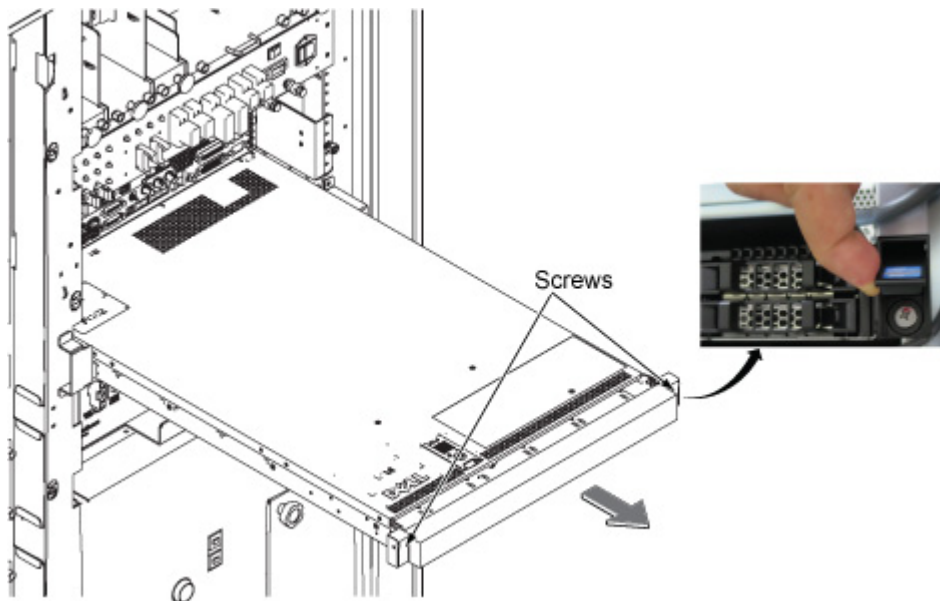
FAILURE TO PROPERLY POWER DOWN THE ICN CAN RESULT IN PREMATURE ICN FAILURE.

Never turn the ICN ON/OFF from the power button on the front. There is software running on ICN disk drives that can be corrupted if not shut down correctly.

Follow the procedure for removing power to the ICN chassis.

1. Open the PGR cabinet front door.
2. Loosen 2 captive screws under the screw cover and slide out the ICN.
3. Slide the ICN out of the cabinet until the ends of the rails touch the stoppers.

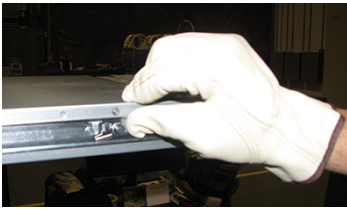
Figure 4-1 Slide the ICN out



4. Disconnect all the cables from the rear of the ICN, including the AC power plug(s). Note the locations for reinstallation.

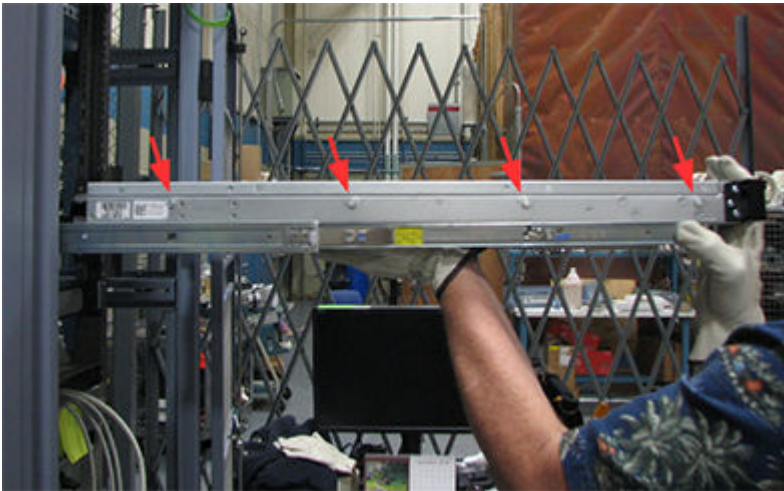
5. Push the two release tabs down and move the ICN forward.

Figure 4-2 Slide rails and release tabs



6. Lift the ICN up to release it from the bracket.
There are four studs on each side of the ICN that fit into the rail.

Figure 4-3 Removing ICN from rails



Chapter 5 Installing the Image Compute Node (ICN)

5.1 Replacing the Image Compute Node (ICN) network cables

1. Remove the Ethernet cable (5488202-2) from port 2 of the Ethernet switch and discard.
2. Remove the Ethernet cable (5488202-3) from port 3 of the Ethernet switch and discard.
3. Install the Ethernet cable (5488202-8) into the Ethernet switch. The other end of this cable will connect to port 4 of the new ICN.

5.2 Installing the ICN

NOTICE



FAILURE TO PROPERLY POWER DOWN THE ICN CAN RESULT IN PREMATURE ICN FAILURE.

Never turn the ICN ON/OFF from the power button on the front. There is software running on ICN disk drives that can be corrupted if not shut down correctly.

Follow the procedure for removing power to the ICN chassis.

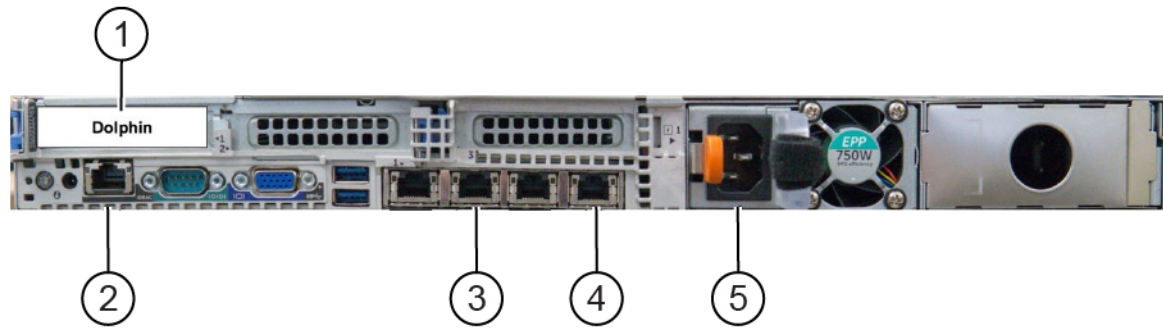
1. With the slide rails out, install the new ICN, lining up the studs on the ICN with the slots on the rails.
2. Connect all other cables that were previously removed to the rear of the new ICN.

NOTE

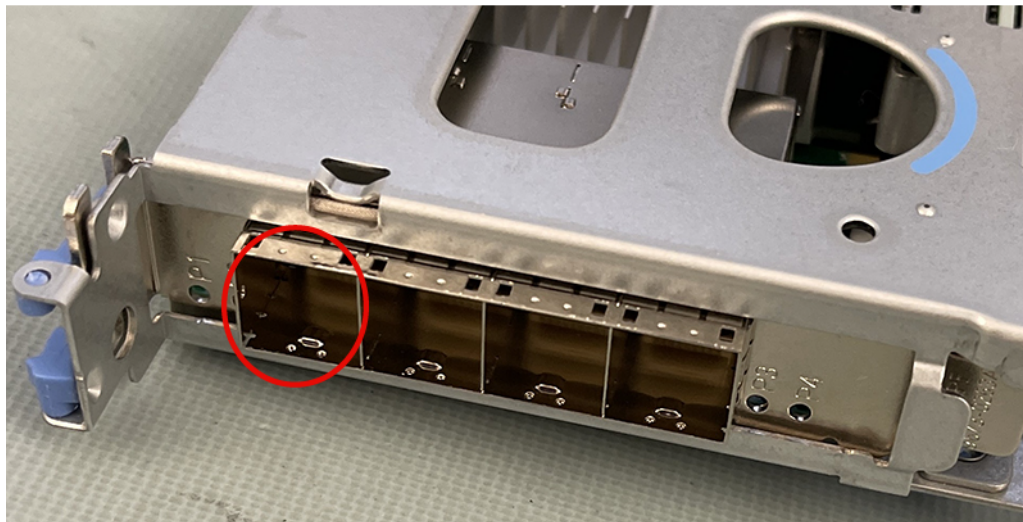
If changing between ICN styles (units with the Dolphin 1 interface with units with the Dolphin 2 interface) make sure the associated PICE cable is available. Some cabinets will contain both cables. Save the alternate cable with the cabinet for future use.

NOTE

Gen 7 and Gen 7-DL ICNs have the same cable connections.

Figure 5-2 ICN cable connections (Gen 7 and Gen 7-DL, Dell R640)

1	PCIe port (Dolphin 1 card), or PCIe P1 port (Dolphin 2 card)
2	iDRAC (management controller for ICN)
3	Raw data server (RDS) - optional, only for research sites
4	Main switch (port 4), only 1 network cable is needed for GEN 7 and Gen 7-DL ICN
5	Power

Figure 5-3 PCIe cable connection - Dolphin 2 card**NOTE**

Ethernet ports 1 & 3 are NOT used. Ethernet port 2 is used ONLY when the system is connected to a Raw Data Server (external to the MR System), it should be left free in all other cases. Not following this configuration may lead to VRE configuration failures and/or boot-up/TPS failures.

3. Push the two release tabs down and move the ICN all the way into the PGR cabinet until it locks in place.
4. Install 2 screws.
5. Restore all cabinet covers.

Chapter 6 Removing LOTO

6.1 Removing LOTO - PGR PDU/Gradient Subsystem

1. Alert necessary personnel that power is being restored.
2. Remove the lock and tag.
3. Switch on the MDP's PDU breaker.
4. Press the green **EMO RESET** button on the PDU.

Figure 6-1 EMO RESET button

5. Power up the host PC.

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Chapter 7 Determining software installation

7.1 Determining software installation

Based on the Gen 7 Image Compute Node (ICN) compatibility, determine the next steps needed. Software Load From Cold (LFC) is required in some cases and other cases only require configuring the ICN. Refer to the below tables to determine the next actions.

Use the below task index in combination with [Table 7-2 Software upgrade steps on page 31](#) to determine the next steps for the upgrade.

Table 7-1 Task index

Task index	Task description	Document reference
1	Replace the ICN	4.1 Removing the ICN on page 25 and 5.2 Installing the ICN on page 27
2	Configure the ICN	8.1 Configuring the ICN - VRE on page 33
3	Upgrade software by performing LFC	Refer to <i>Installing host system software (quick start)</i> from the online documentation library.

The below table shows the next steps to be performed based on the task number per index.

Table 7-2 Software upgrade steps

Upgrade To ---->	Gen 7 ICN, SW 28	Gen 7 DL ICN, SW 28	Gen 7 ICN, SW 29*	Gen 7 DL ICN, SW 29*
Upgrade From				
Gen 5 and earlier ICN	Not Applicable	Not Applicable	1, 3	1, 3
Gen 6 ICN, SW 28	1, 2	Not Compatible	1, 3	1, 3
Gen 6 ICN, SW 29	Not Applicable	Not Applicable	1, 2	1, 2
Gen 7 ICN, SW 28	Not Applicable	Not Applicable	3	1, 3
Gen 7 ICN, SW 29	Not Applicable	Not Applicable	Not Applicable	1, 2

* Gen 7 ICN with the Dolphin 2 Interface requires a software level of DV29.1_R02_SP01 or DV29.0_R02_SP01 or later.

If the ICN is being upgraded to the Gen7-DL as part of the Air Recon DL option upgrade, make sure the airrcndl2d option key is installed and the system is rebooted for the key to initialize the Air Recon DL protocols. For more information refer to *Air Recon DL operation and FE troubleshooting theory* in the online documentation library.

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Chapter 8 Configuring the Image Compute Node (ICN) - VRE

8.1 Configuring the ICN - VRE

Use this procedure if Load From Cold (LFC) with the required software has already been completed.

The following procedure describes manual ICN-VRE configuration. ICN compatibility is dependent on software revision. Refer to *PGR Cabinet* for compatibility. Volume Recon Engine (VRE) and Image Compute Node (ICN) are terms that are used interchangeably. VRE configuration installs the software onto the ICN. This procedure is needed after ICN replacement. This procedure may also be needed when troubleshooting issues with the ICN system. Refer to *Troubleshooting the Image Compute Node (ICN)* for more information.

1. **NOTE**

If you are replacing the ICN, power on the system and the host PC, and log in as root.

NOTE

(For DV29.0 or later) The root password is no longer **operator**. Passwords are managed according to *MR Cyber Security Manual* in the online documentation library.

Click **Guided Install** and then click **Start**.

2. Invoke Guided Install as root.

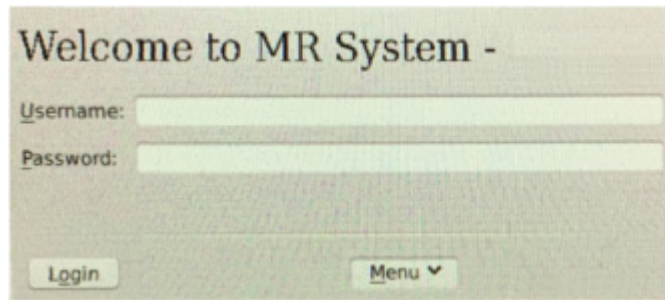
- a. When you power up the host computer and see the logon screen, click on **root login**.

Figure 8-1 Logon screen



- b. Type the user name: **root** and type the root password.

Figure 8-2 Welcome logon screen



NOTE

It is possible that the customer changed the default password. If you cannot log on , contact the customer for the correct password.

Logging on as the root user automatically shows the Linux desktop and launches the *Guided Install Starter* window.

- c. Click **Yes** to invoke the guided install.

NOTE

If **No** has already been selected, right-click on the Linux desktop and select **Start Guided Install**.

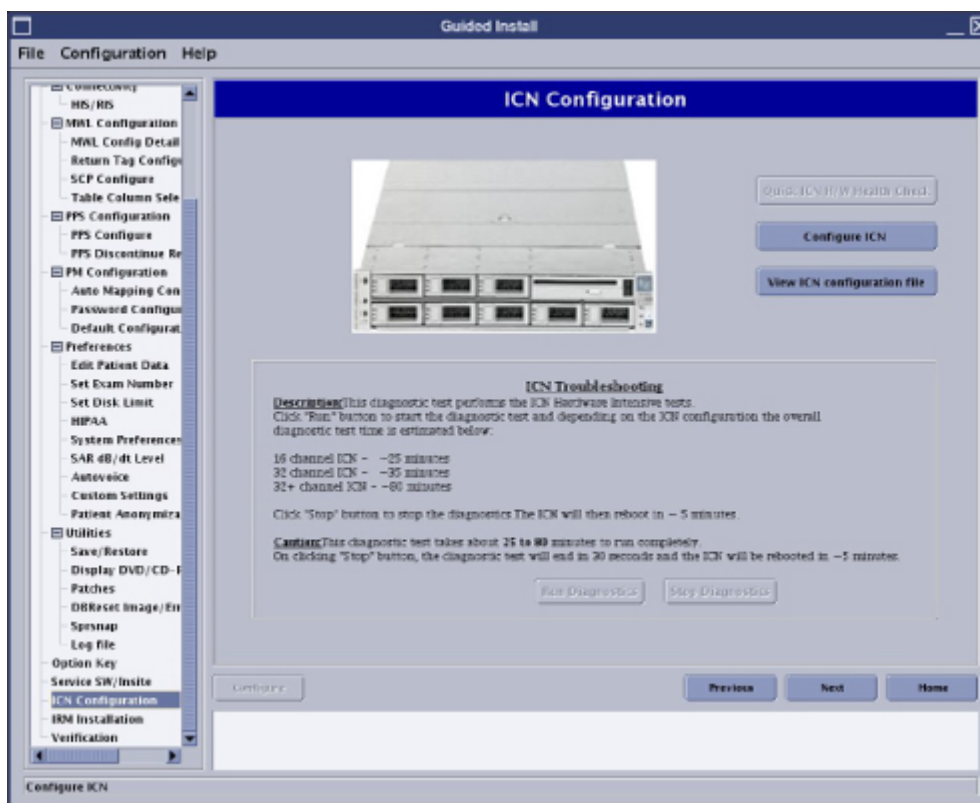
3. Select **ICN Configuration**, click **Configure ICN**, and click **OK**.
4. On the ICN configuration screen, click **View VRE Configuration file** to view the output status of the ICN.

NOTE

Information will only be shown if ICN was previously configured in the software. Note the serial number and check the output to verify the new serial number.

- Click **Configure ICN**.

Figure 8-3 Guided install – ICN configuration

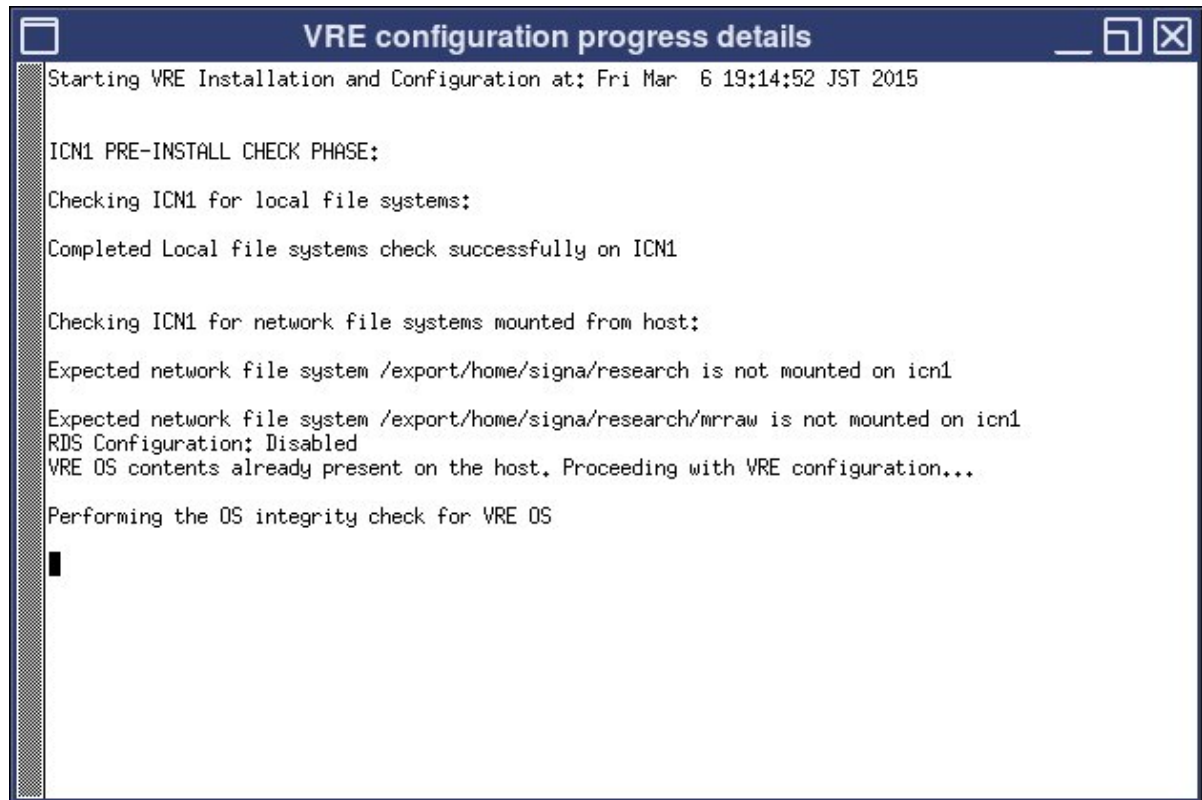


- If prompted, insert the OS DVD, or USB as applicable, that came with the system software into the host PC and click **OK** to continue the ICN configuration (this step is only necessary if repository files are corrupted on the host computer).

ICN configuration progress and details appear in a separate window. At the end of ICN configuration, another dialog box is displayed with ICN configuration success or failure message.

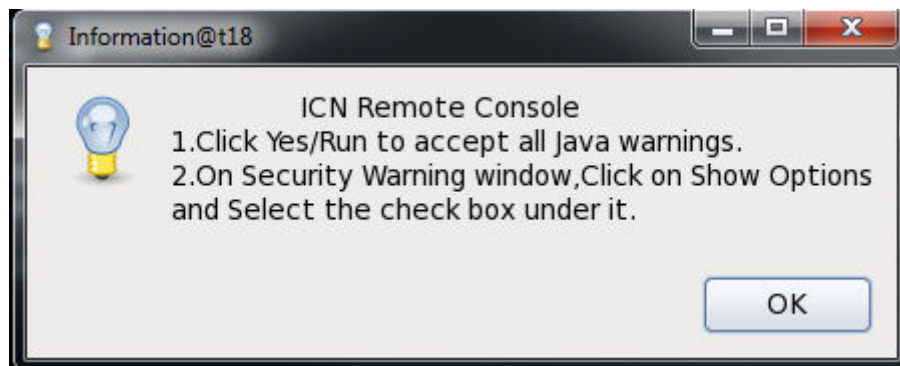
7. If ICN configuration fails, review the dialog box and details in the separate window to determine next steps.

Figure 8-4 ICN progress details



8. When the **ICN Remote Console** window appears, click **OK**, and then **Continue**. This will install the Java security options.

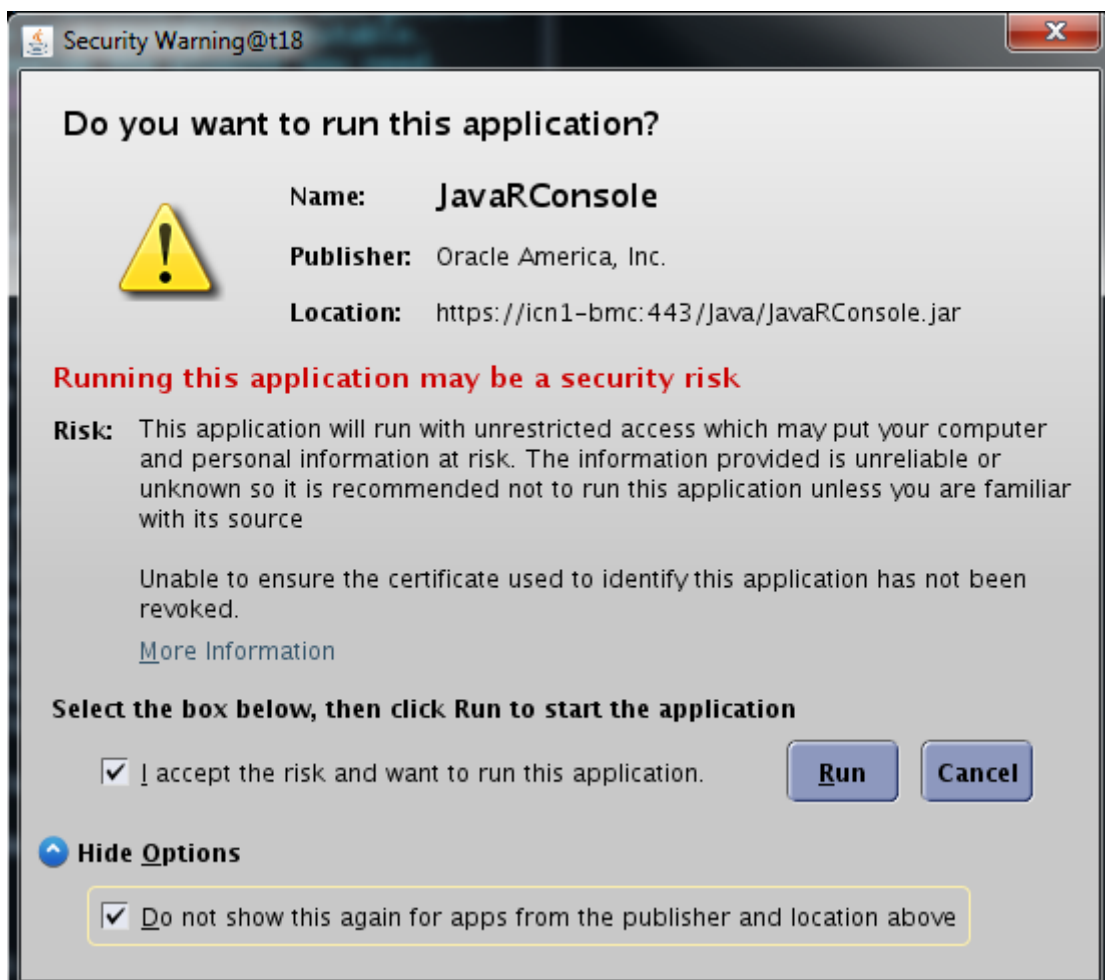
Figure 8-5 ICN remote console



9. Once the *Java Security Warning* window appears select the following:
 - a. Select the **I accept the risk and want to run this application** checkbox.
 - b. Click on **Show Options** and select the **Do not show this again for apps from the publisher and the location above.** checkbox.

- a. Click **Run**.

Figure 8-6 Security Warning



The *Remote Console* window opens showing the progress of the installation.

10. At the end of VRE configuration, a dialog box displays with either a success or failure message. If VRE configuration fails, review the dialog box message and details in the separate window to determine next steps.

NOTE

On the ICN configuration screen, you can click **View VRE Configuration file** to view the output status of the ICN.

11. Click **OK** twice for the VRE configuration success and ICN configuration success messages.

NOTE

A reboot is required after complete ICN configuration.

12. Remove the OS DVD or USB SW installer.

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Chapter 9 Doing a check scan

9.1 Doing a check scan

Make sure that you use the DQA phantom, so that the image can be analyzed for correct orientation.

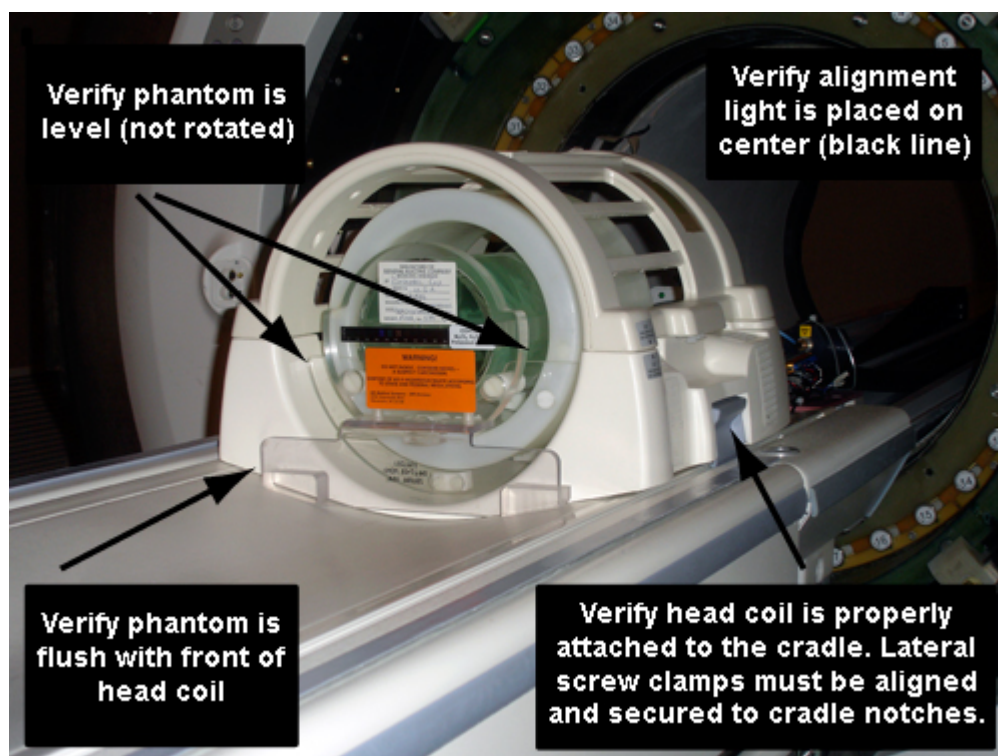
NOTICE



Depending on site configuration and/or rules, you may want to temporarily disable Auto Archive or Auto Network.

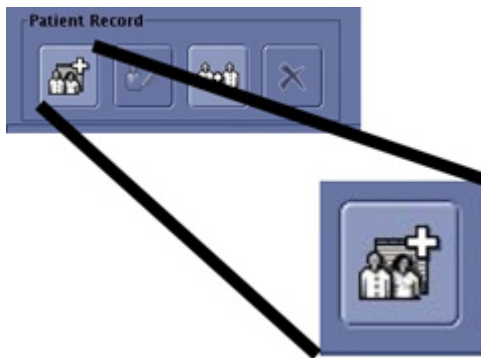
1. Attach the head coil to the cradle and position the DQA phantom inside the coil.
2. Landmark on the center line of the phantom. The laser must be in the middle of the line. Press **Advance to Scan**.

Figure 9-1 Phantom orientation, split-top head coil



- Click the **New Worklist** icon.

Figure 9-2 New worklist icon



- Select a **Patient ID** workflow (other than **geservice**).

Figure 9-3 Entering a new work list

The screenshot shows a software interface for entering a new work list. At the top, there's a 'Patient Record' section with icons for adding, editing, and deleting records. Below this is a table with columns: Scheduled, Worklist, Name, Patient ID, Accession, SPS Description, Modality, Ref. Physician, Scheduled, Status, Local Protocol, Recomm., and Caution. The first row shows a patient scheduled for 01:58:31 PM, Local, with a Patient ID of 000000, Accession of 000000, SPS Description of MR, Modality of MR, Ref. Physician of 11-26-2007, Status of Started, and Local Protocol of 30.11 3-Plane 2D Localizer.

Below the table is a form with three main sections: Patient, Exam, and Other Information.

Patient Section:

- Name: Last Name (), First Name ()
- * Patient ID: A00000
- Scheduled Date: 11-26-2007
- Date Of Birth: MM-DD-YYYY ()
- Age: Year (), Month (), Weeks (), Days ()
- Sex: ()
- * Weight: Lbs (111.0), Kgs (50.35)

Exam Section:

- Ref. Physici...: Last Name (), First Name ()
- Accession: ()
- SPS Description: ()
- Exam Description: ()
- Procedure ID: ()
- Operator's Initial: ()
- Radiologist: ()
- Protocol: 30.11 3-Plane 2D Localizer
- Favorite Protocols: Select Favorite ()
- Show All Protocols... ()

Other Information Section:

- This is patient sensitive information, please adhere to patient confidentiality rules. Please verify the information using either the patient's medical record or directly with the patient to guarantee its accuracy.
- Allergies: NONE ()
- Pre-Med: NONE ()
- Pregnancy Status: No Entry ()
- History: ()

At the bottom right, there is a 'Reopen Exam' button.

- Enter for weight: **111**.
- Click **Show All Protocols...**
- When the **Protocol Selector** menu opens, click **Protocol Details**.

8. Select the Protocol Library: **GE**.

Figure 9-4 Setting protocols



9. Select **Adult**.
10. Click the **Template** tab.

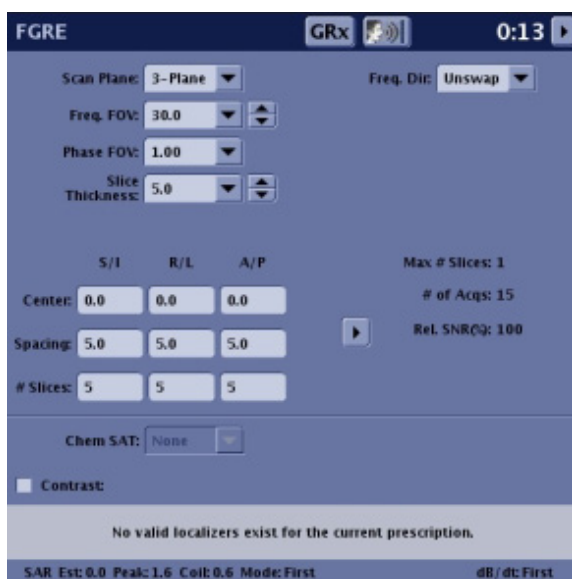
11. Select **3 - Plane 2D Localizer** and click the arrow to move the selection to the box on the right.

Figure 9-5 Protocol selected



12. Click **Accept**.
13. Click **Save**.
14. Click **Start Exam**.
15. Enter the following image parameters.

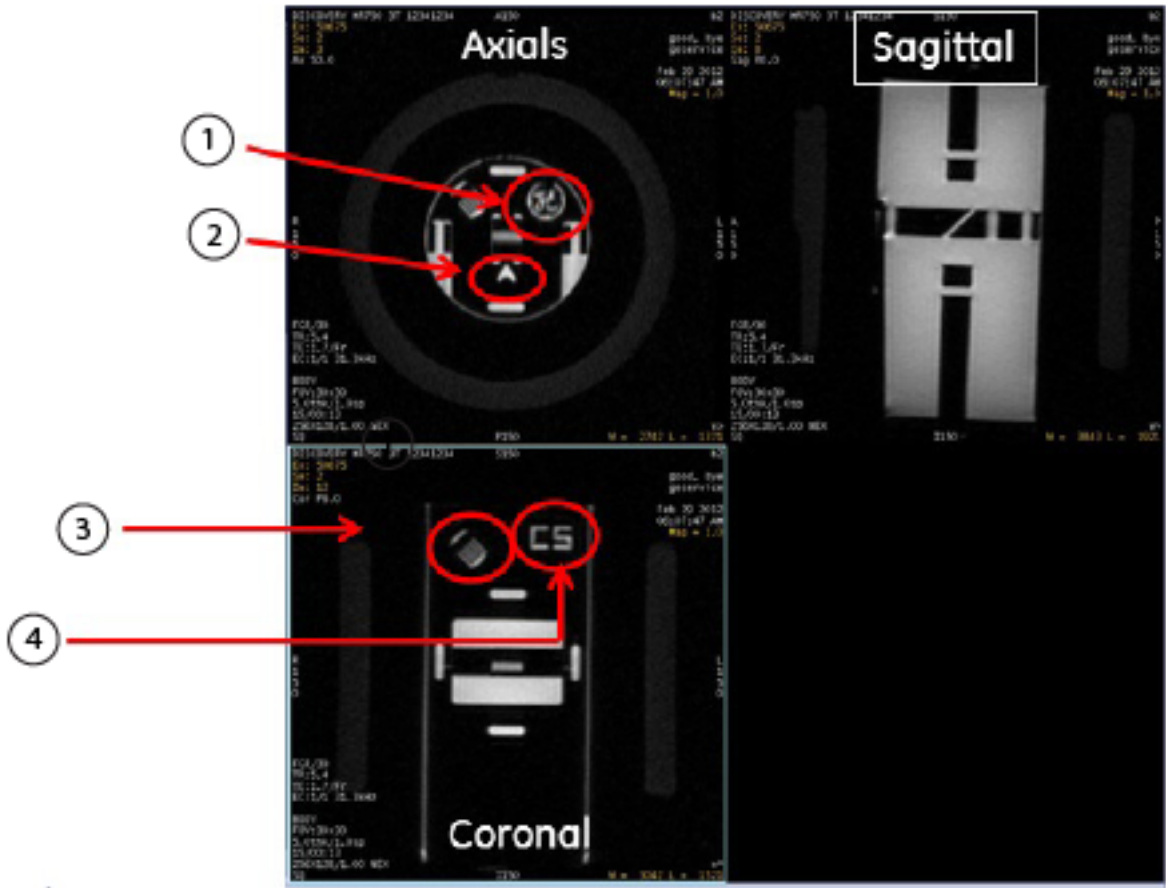
Figure 9-6 Image parameters



Option	Description
Frequency field of view (FOV) (Freq. FOV)	30.0
Phase FOV	1.00
Slice Thickness	5.0

- 16. Click **Save RX**.
- 17. Click **Scan**.
- 18. Review the images to confirm that the orientation is correct.

Figure 9-7 Check scan image orientation



Axials:	
1	GE logo in upper right corner
2	A is displayed
Coronal & Sagittal:	
3	Comb in upper left corner
4	CS displayed in upper right corner

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